Desarrollo de Aplicaciones Híbridas - OPT

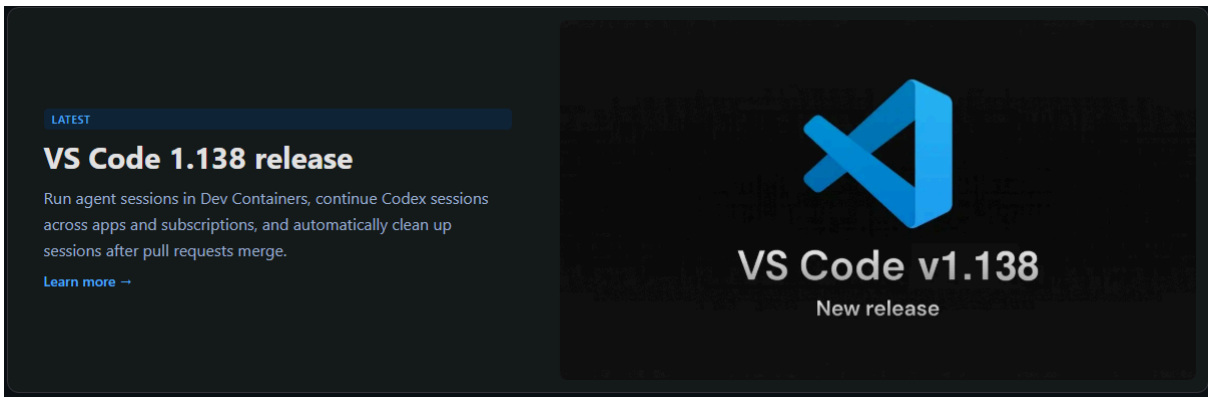
Instalación del Entorno & Primera App

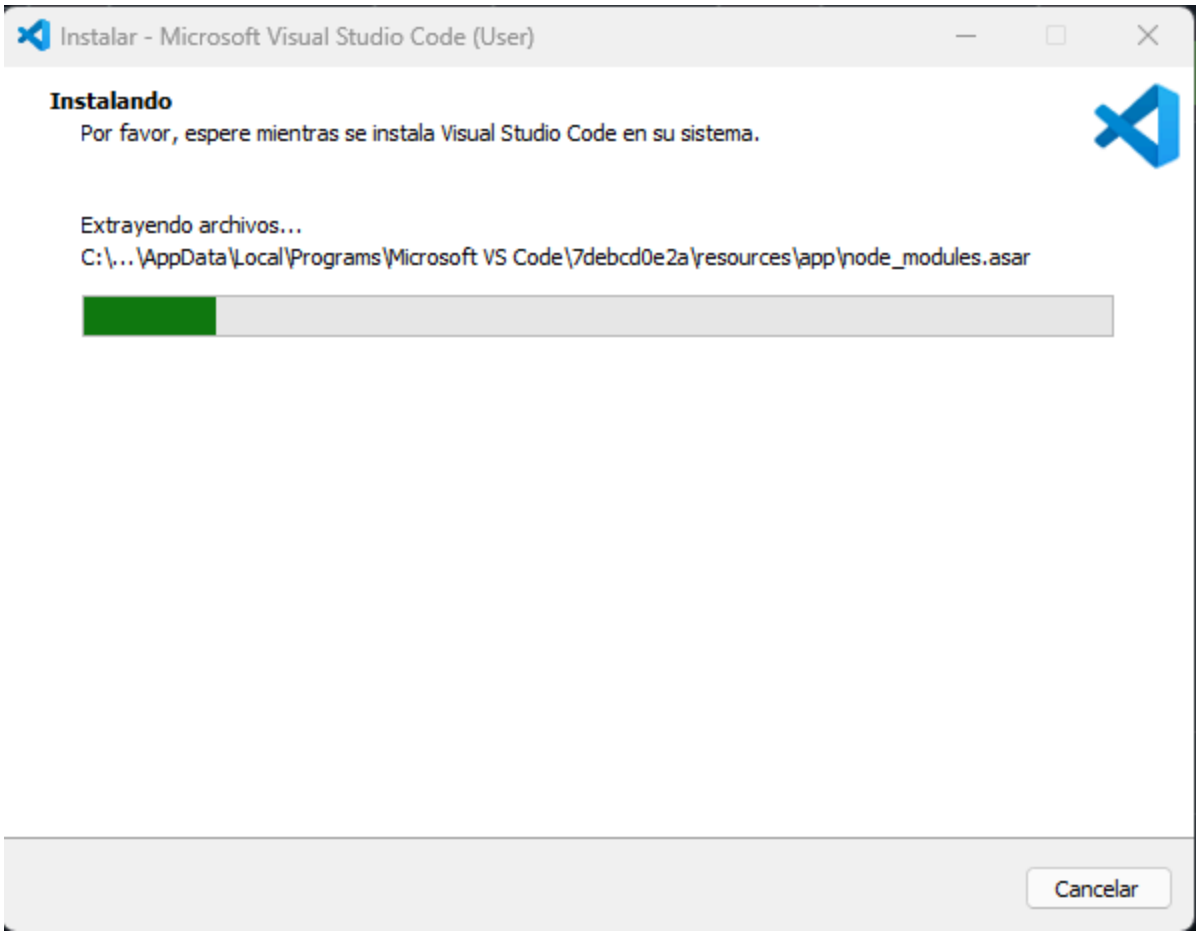
PABLO BOCANEGRA CORRALES



Editor de Código: Visual Studio Code

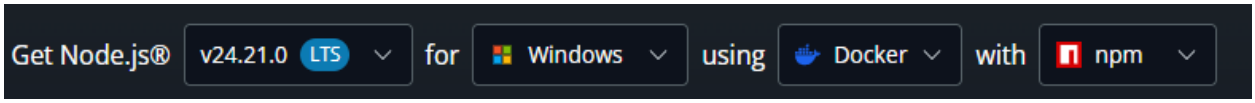
Descargamos desde la página oficial la versión mas reciente: v1.138



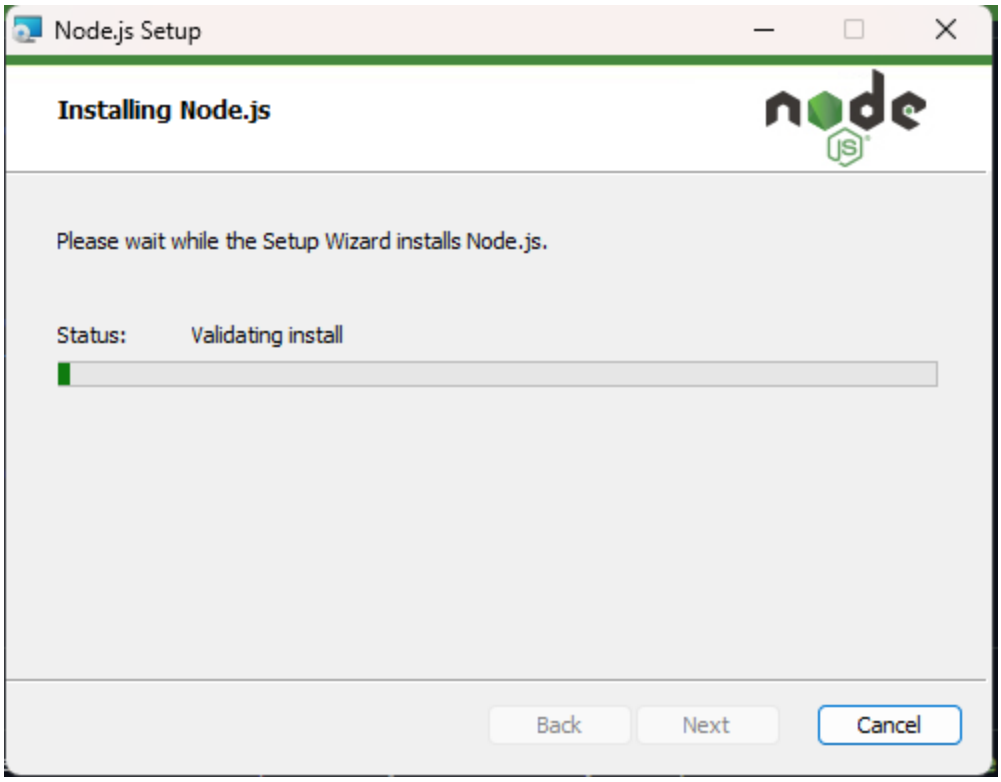


Entorno de Ejecución: Node.js y npm

Al igual que con VisualStudio, descargamos el software desde la página oficial la versión v24.21.0 (LTS).



Usando el archivo .msi, un paquete preparado de fábrica para instalar con el Wizard haremos la instalación sobreescribiendo los archivos de la versión antigua que tenia instalada.



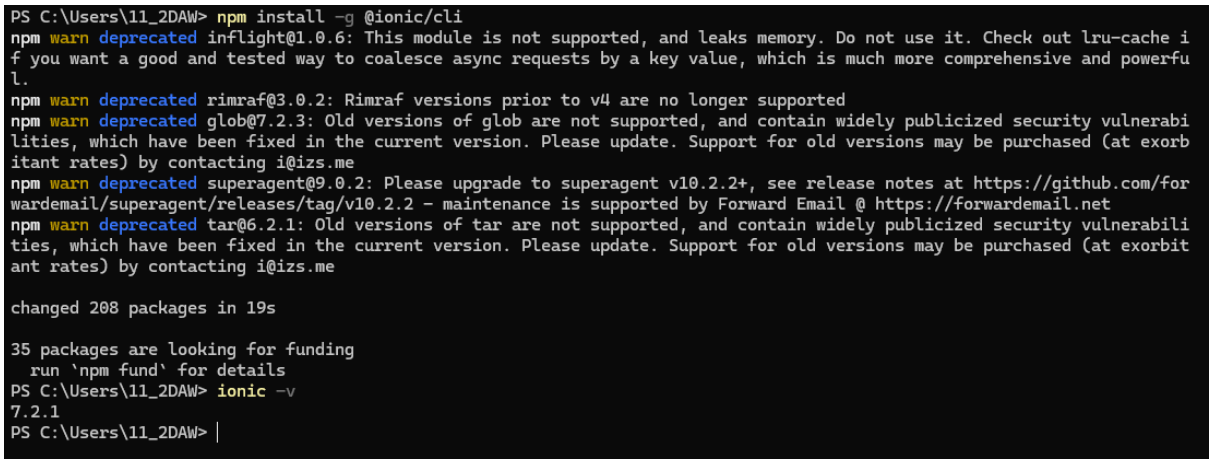
Para comprobar la versión instalada simplemente nos dirigimos a la terminal y escribimos el nombre del paquete -v: "*npm -v*" "*node -v*"

```
PS C:\Users\11_2DAW> node -v
v24.21.0
PS C:\Users\11_2DAW> npm -v
11.19.0
```

Interfaz de Línea de Comandos (CLI) de Ionic

Desde la terminar de Windows, usando el siguiente comando instalaremos el software de Ionic, "*npm install -g @ionic/cli*"

Y para comprobar que la versión es la correcta, usaremos el comando "*ionic -v*"



Creación de Proyectos con Ionic

Una vez todas las herramientas instaladas, haremos la creación de nuestra primera aplicación. Lo primero que haremos será crear un directorio donde almacenaremos los proyectos y crearemos nuestra primera app con la plantilla **tabs**.

```
PS C:\Users\11_2DAW> cd .\Desktop\
PS C:\Users\11_2DAW\Desktop> mkdir dah_2627


    Directorio: C:\Users\11_2DAW\Desktop


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        18/09/2026      9:20                dah_2627


PS C:\Users\11_2DAW\Desktop> cd .\dah_2627\
PS C:\Users\11_2DAW\Desktop\dah_2627> ionic start mi-primera-app tabs
```

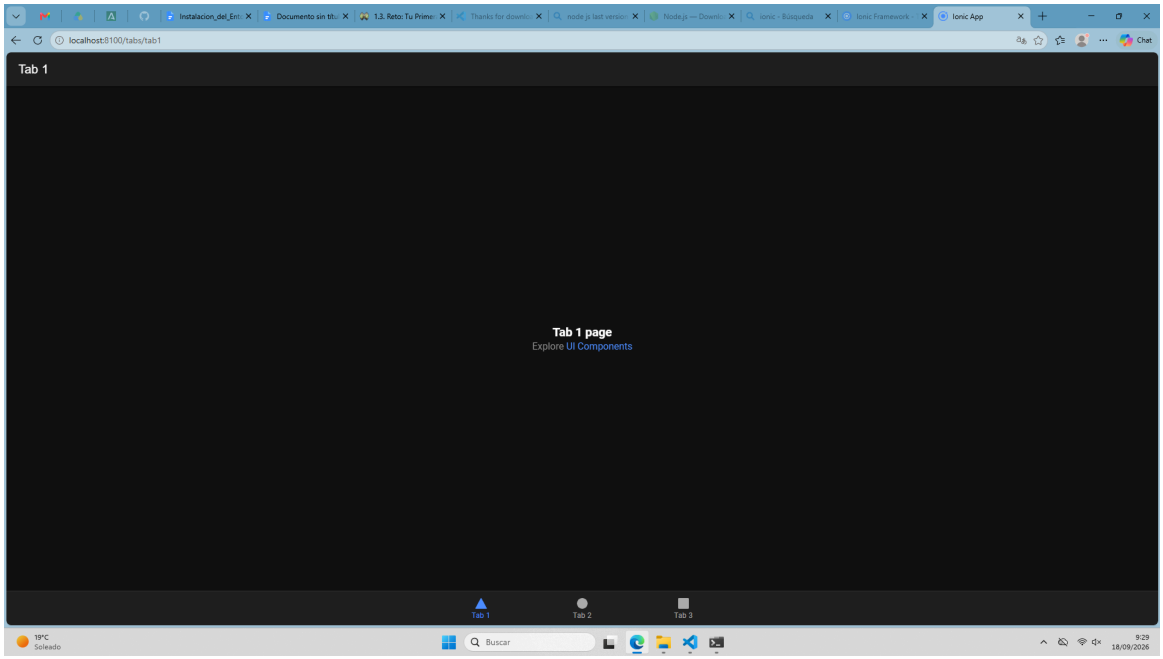
Marcaremos las siguientes opciones para seguir con la instalación

```
? Framework: Angular
? Would you like to build your app with Standalone Components or NgModules?
 Standalone components are the default way to build with Angular that simplifies the way you build your app.
 To learn more, visit the Angular docs:
 https://angular.dev/guide/components

 Standalone
√ Preparing directory .\mi-primera-app in 1.29ms
√ Downloading and extracting tabs starter in 2.07s
```

Una vez la app montada, el siguiente paso es lanzarla con el comando ionic serve desde el directorio donde se encuentre.

```
PS C:\Users\11_2DAW\Desktop\dah_2627> cd .\mi-primera-app\
PS C:\Users\11_2DAW\Desktop\dah_2627\mi-primera-app> ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
[ng] › Building...
```



En VisualStudioCode, nos dirigimos a la carpeta \mi-primera-app\src\app\tab1 para modificar la página “tab1”

tab1.page.html

src > app > tab1 > tab1.page.html > ...

```
<ion-header [translucent]="true">
  <ion-toolbar>
    <ion-title>
      Tab 1
    </ion-title>
  </ion-toolbar>
</ion-header>

<ion-content [fullscreen]="true">
  <ion-header collapse="condense">
    <ion-toolbar>
      <ion-title size="large">Tab 1</ion-title>
    </ion-toolbar>
  </ion-header>

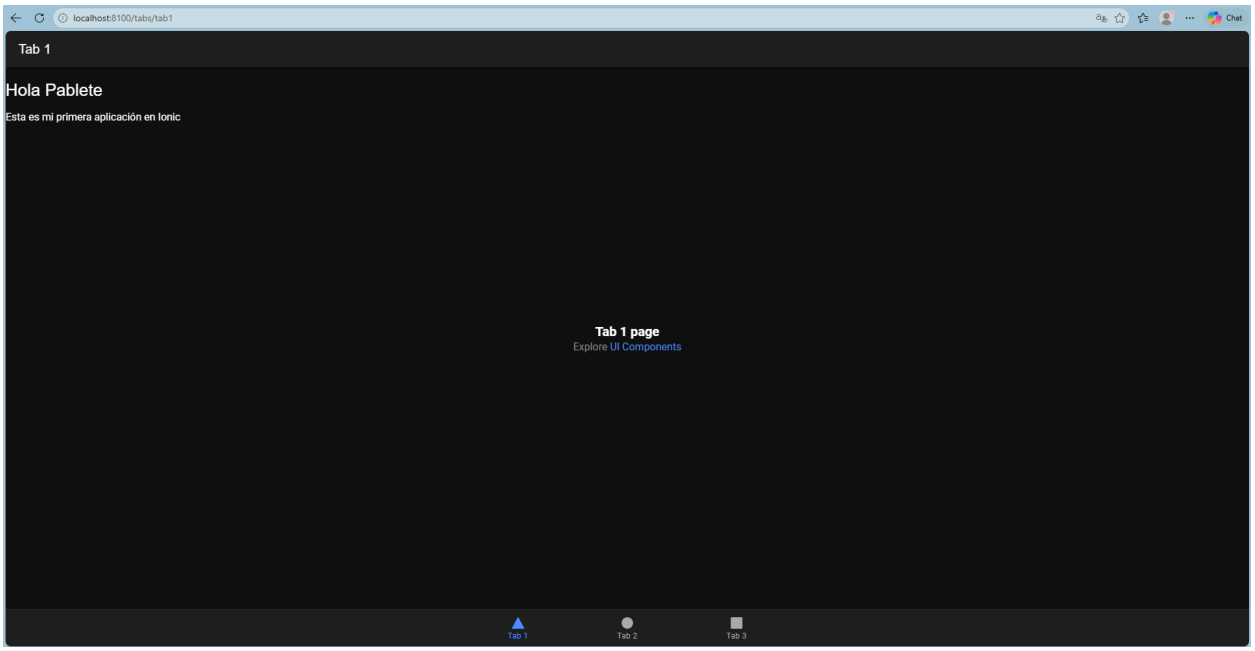
  <app-explore-container name="Tab 1 page"></app-explore-container>

  <h1> Hola Pablete</h1>
  <p> Esta es mi primera aplicación en Ionic</p>


</ion-content>

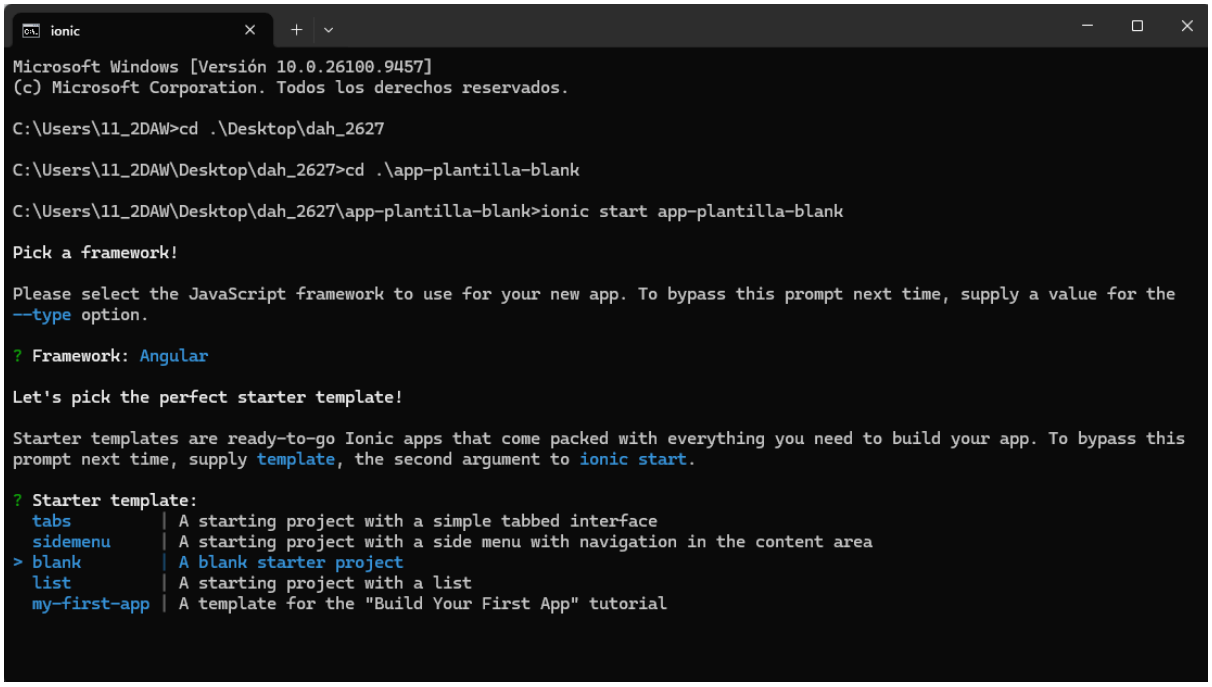
```

Guardamos cambios, recargamos la página y observamos que se han actualizado bien

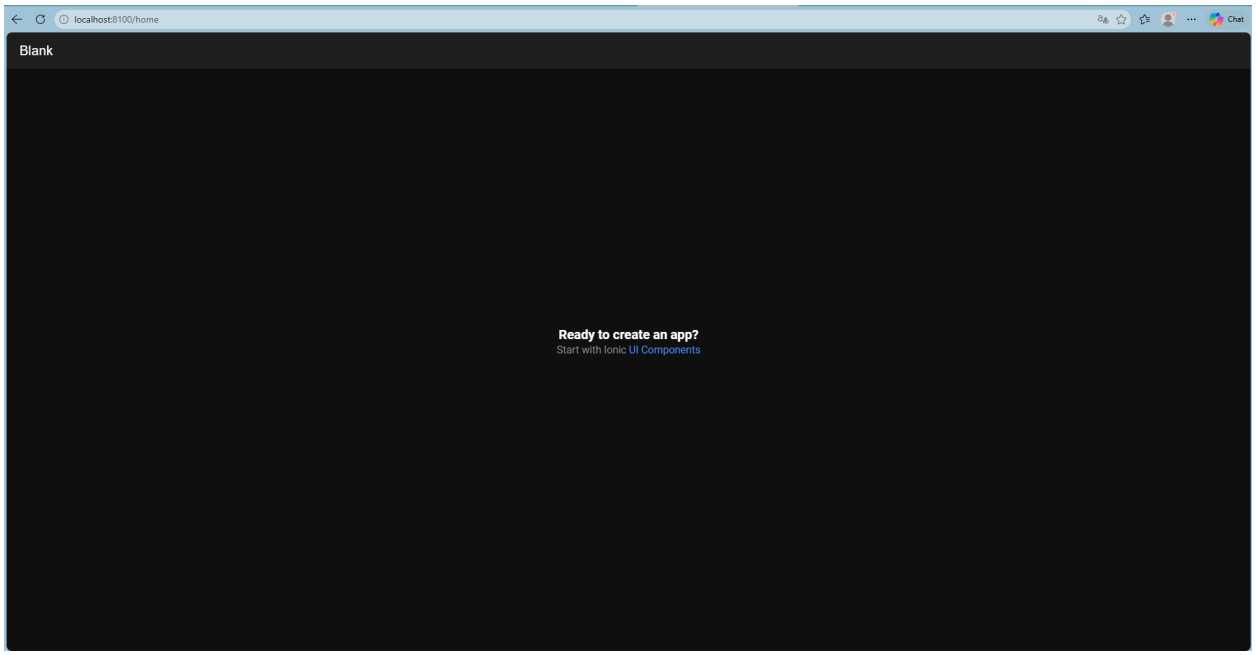


Plantilla “Blank”

Crearemos una app por cada plantilla que tenga el selector, empezando por la plantilla “blank” para la cual he creado un directorio llamado “app-plantilla-blank” al igual que para el resto de plantillas.







Plantilla “SideMenu”

```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-sidemenu>ionic start app-plantilla-sidemenu

Pick a framework!

Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.

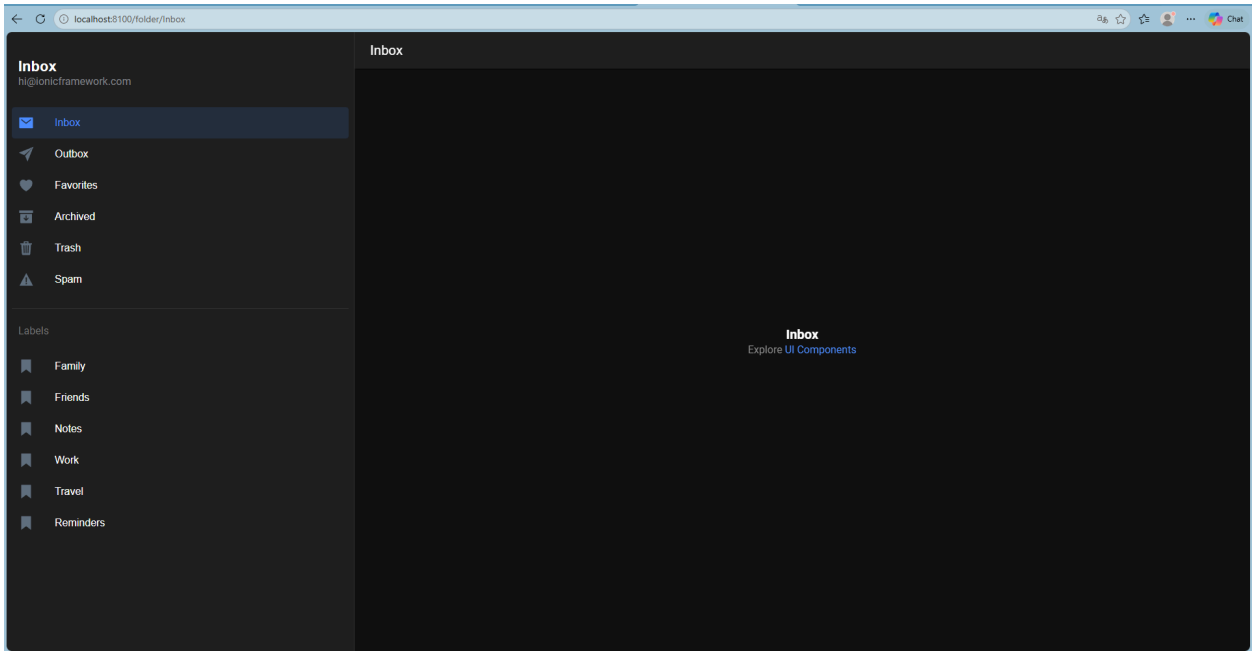
? Framework: Angular

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.

? Starter template:
  tabs         | A starting project with a simple tabbed interface
> sidemenu     | A starting project with a side menu with navigation in the content area
  blank        | A blank starter project
  list         | A starting project with a list
  my-first-app | A template for the "Build Your First App" tutorial
```

```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-sidemenu\app-plantilla-sidemenu>ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
```



Plantilla “List”

```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-list>ionic start app-plantilla-list

Pick a framework!

Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.

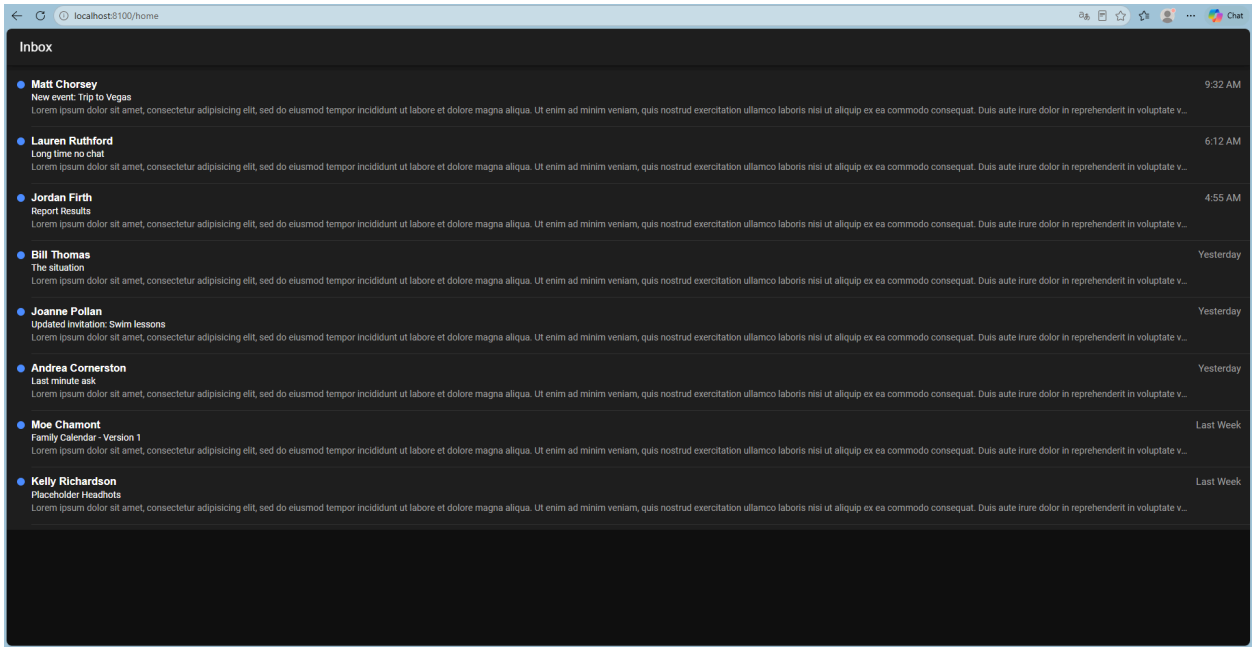
? Framework: Angular

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.

? Starter template:
  tabs          | A starting project with a simple tabbed interface
  sidemenu      | A starting project with a side menu with navigation in the content area
  blank         | A blank starter project
> list          | A starting project with a list
  my-first-app  | A template for the "Build Your First App" tutorial
```

```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-list\app-plantilla-list>ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
```



Plantilla “my-first-app”

```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-first>ionic start app-plantilla-first

Pick a framework!

Please select the JavaScript framework to use for your new app. To bypass this prompt next time, supply a value for the
--type option.

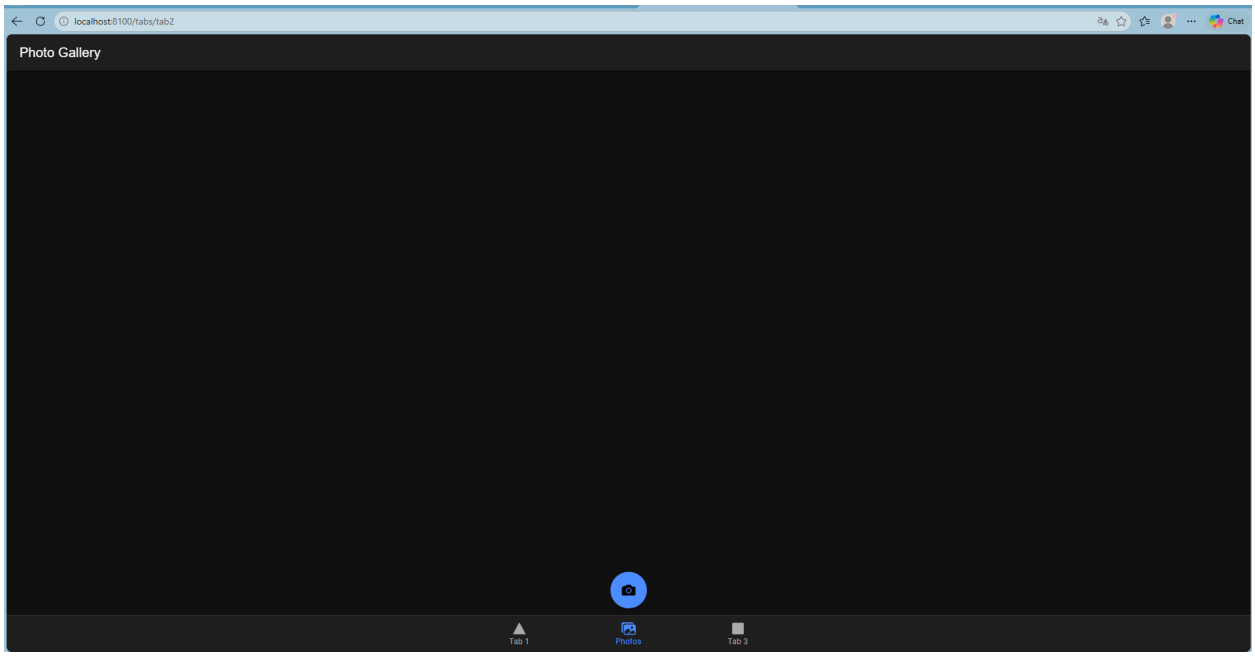
? Framework: Angular

Let's pick the perfect starter template!

Starter templates are ready-to-go Ionic apps that come packed with everything you need to build your app. To bypass this
prompt next time, supply template, the second argument to ionic start.

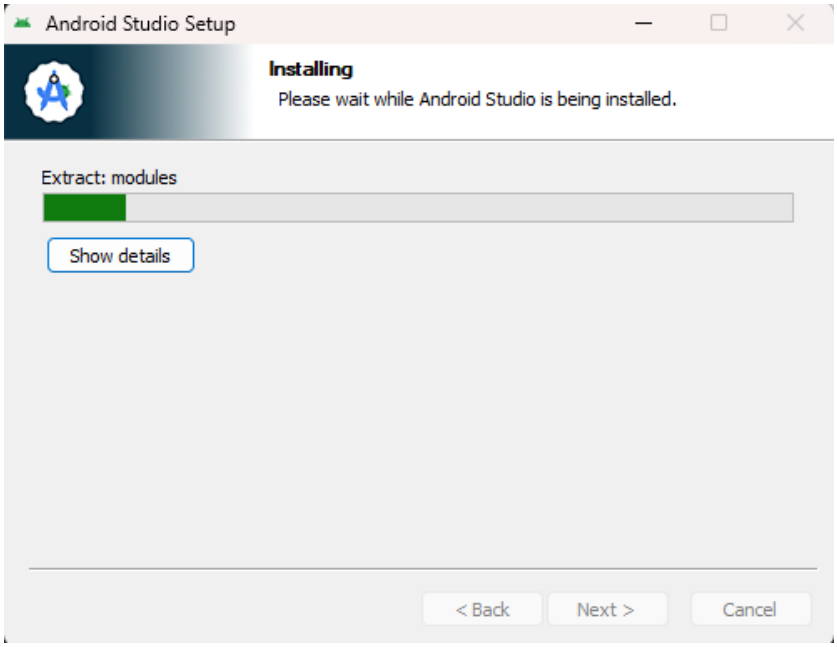
? Starter template:
  tabs          | A starting project with a simple tabbed interface
  sidemenu      | A starting project with a side menu with navigation in the content area
  blank         | A blank starter project
  list          | A starting project with a list
> my-first-app  | A template for the "Build Your First App" tutorial
```

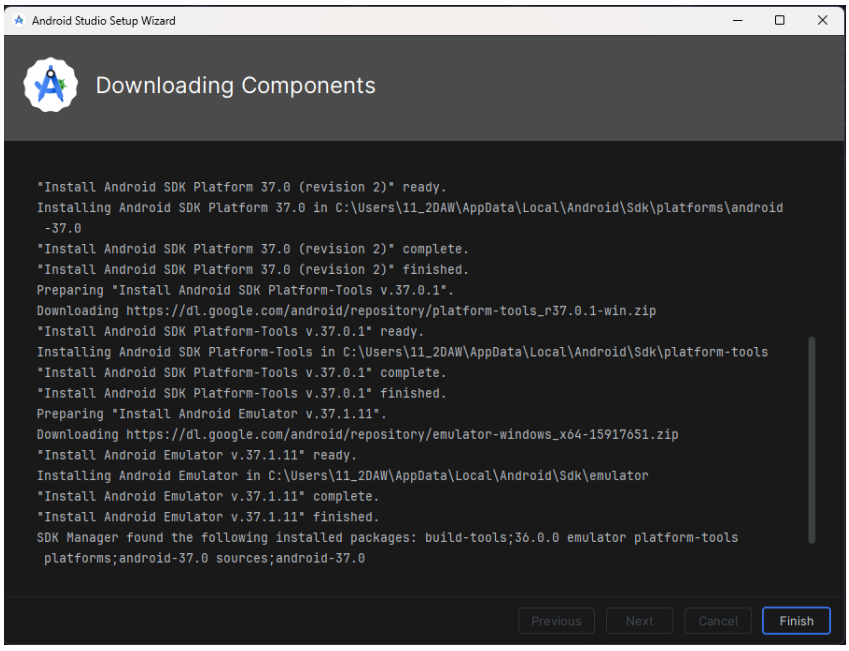
```
C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-first\app-plantilla-first>ionic serve
> ng.cmd run app:serve --host=localhost --port=8100
```



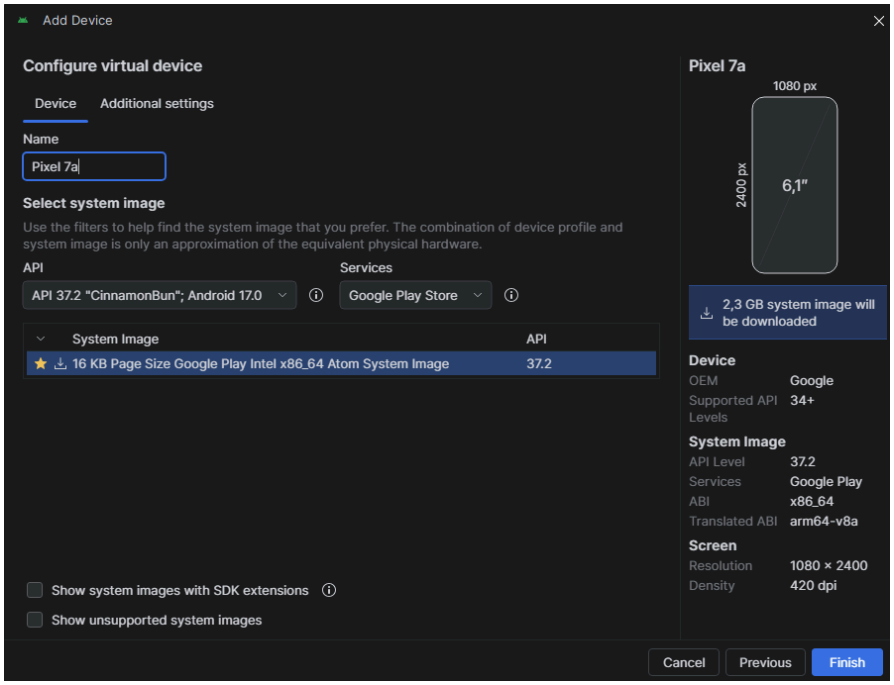
Instalación y Preparación del Entorno Android Studio

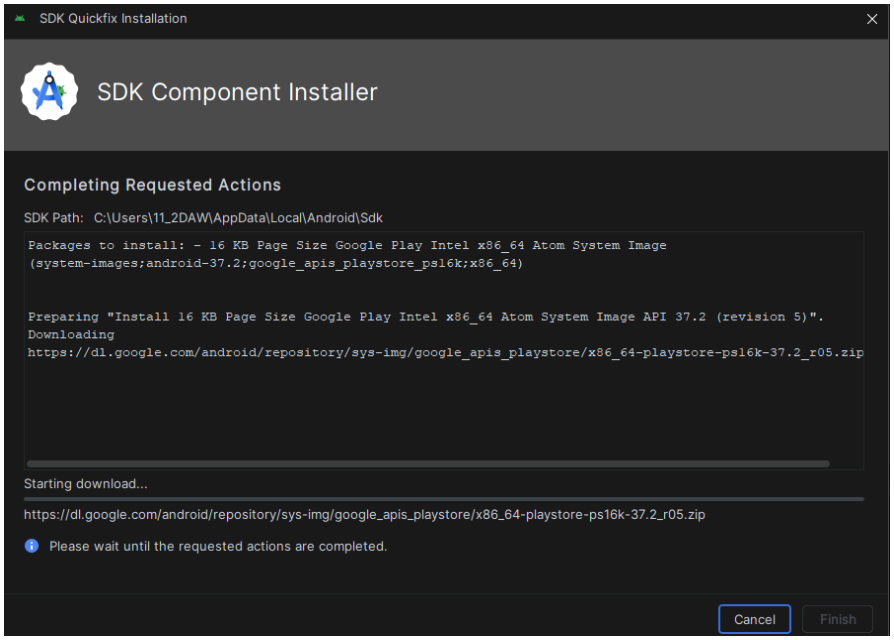
Instalaremos el Android Studio con ayuda del Wizard y elegiremos las opciones estándar para instalarlo

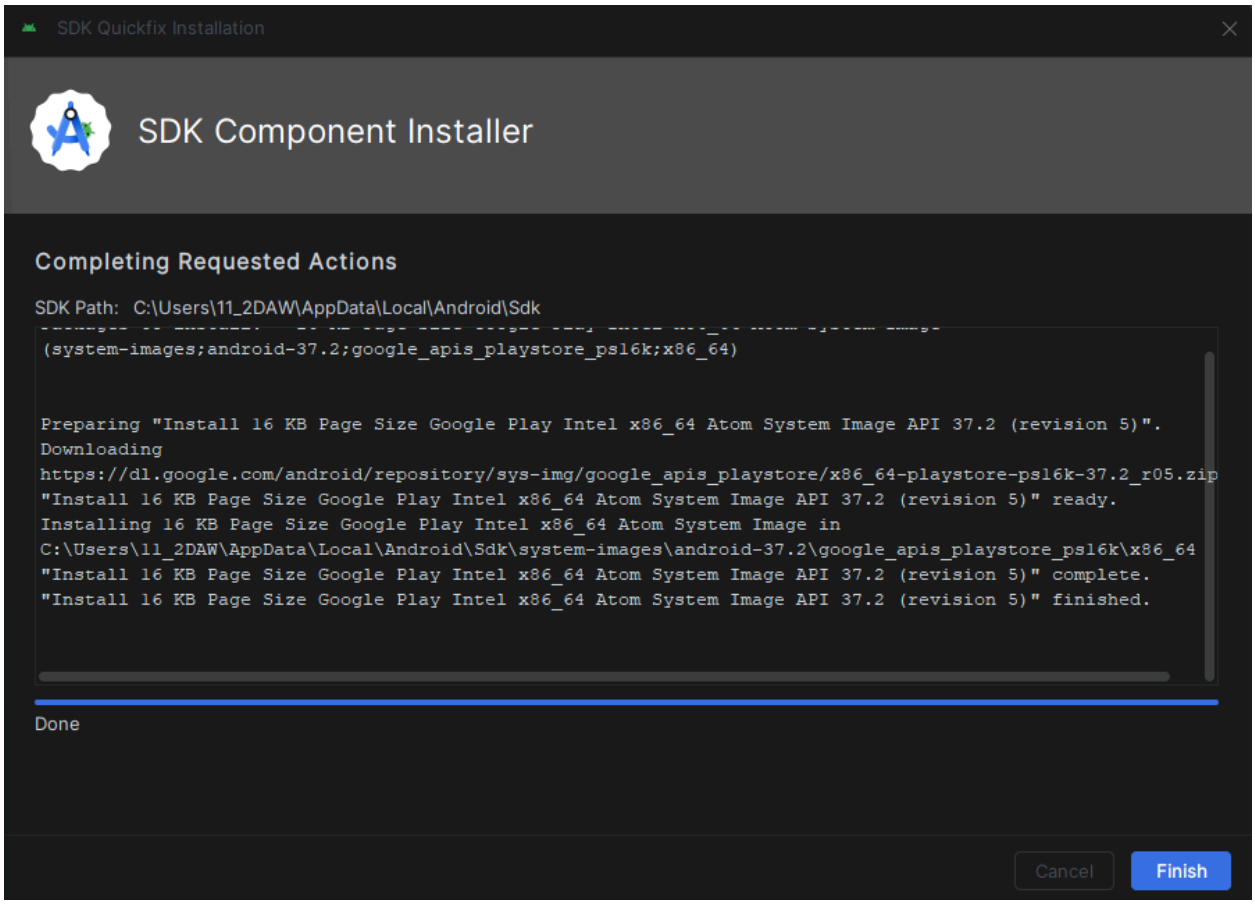




Una vez el entorno instalado crearemos nuestro primer entorno, el cual será un Pixel 7a y dejaremos las opciones por defecto también






SDK Quickfix Installation
SDK Component Installer
Completing Requested Actions
SDK Path: C:\Users\11_2DAW\AppData\Local\Android\Sdk
(system-images;android-37.2;google_apis_playstore_ps16k;x86_64)
Preparing "Install 16 KB Page Size Google Play Intel x86_64 Atom System Image API 37.2 (revision 5)".
Downloading
https://dl.google.com/android/repository/sys-img/google_apis_playstore/x86_64-playstore-ps16k-37.2_r05.zip
"Install 16 KB Page Size Google Play Intel x86_64 Atom System Image API 37.2 (revision 5)" ready.
Installing 16 KB Page Size Google Play Intel x86_64 Atom System Image in
C:\Users\11_2DAW\AppData\Local\Android\Sdk\system-images\android-37.2\google_apis_playstore_ps16k\x86_64
"Install 16 KB Page Size Google Play Intel x86_64 Atom System Image API 37.2 (revision 5)" complete.
"Install 16 KB Page Size Google Play Intel x86_64 Atom System Image API 37.2 (revision 5)" finished.
Done
Cancel
Finish

Conseguimos iniciar el smartphone y prepararle los requisitos necesarios para iniciar aplicaciones



Cuando ejecutamos el comando para iniciar la app en el movil nos va a pedir que especefiquemos el movil que vamos a usar, en mi caso será el pixel 7a

```
PS C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-list\app-plantilla-list> ionic cap run android
> capacitor.cmd add android
[capacitor] √ Adding native android project in android in 199.88ms
[capacitor] √ add in 200.55ms
[capacitor] [warn] sync could not run--missing www directory.
[capacitor] [success] android platform added!
[capacitor] Follow the Developer Workflow guide to get building:
[capacitor] https://capacitorjs.com/docs/basics/workflow
? Which device would you like to target? Pixel 7a (emulator) (Pixel_7a)
> ng.cmd run app:build
```

Al finalizar la ejecución ha finalizado con este error, asique he buscado y encontrado unos parámetros para el comando que puede hacer que funcione

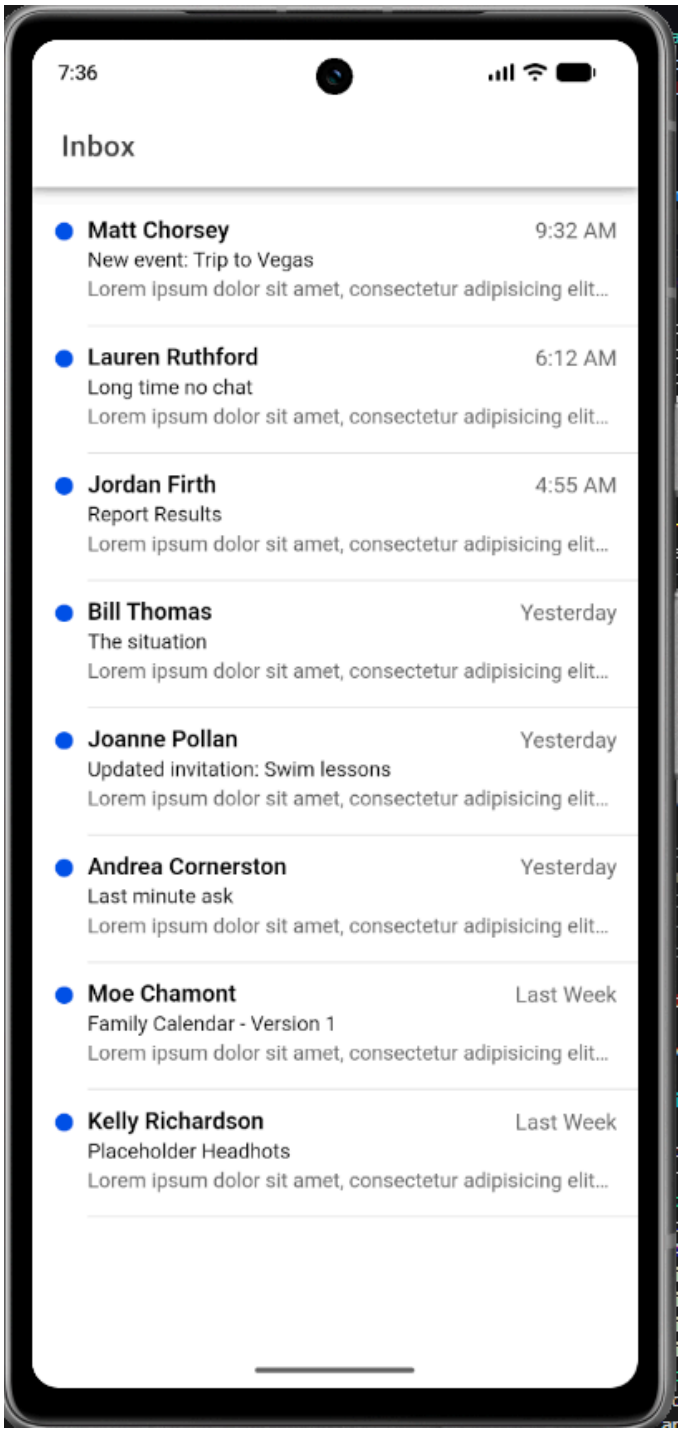
```
[capacitor]
[ERROR] An error occurred while running subprocess capacitor.

        capacitor.cmd run android --target Pixel_7a exited with exit code 1.

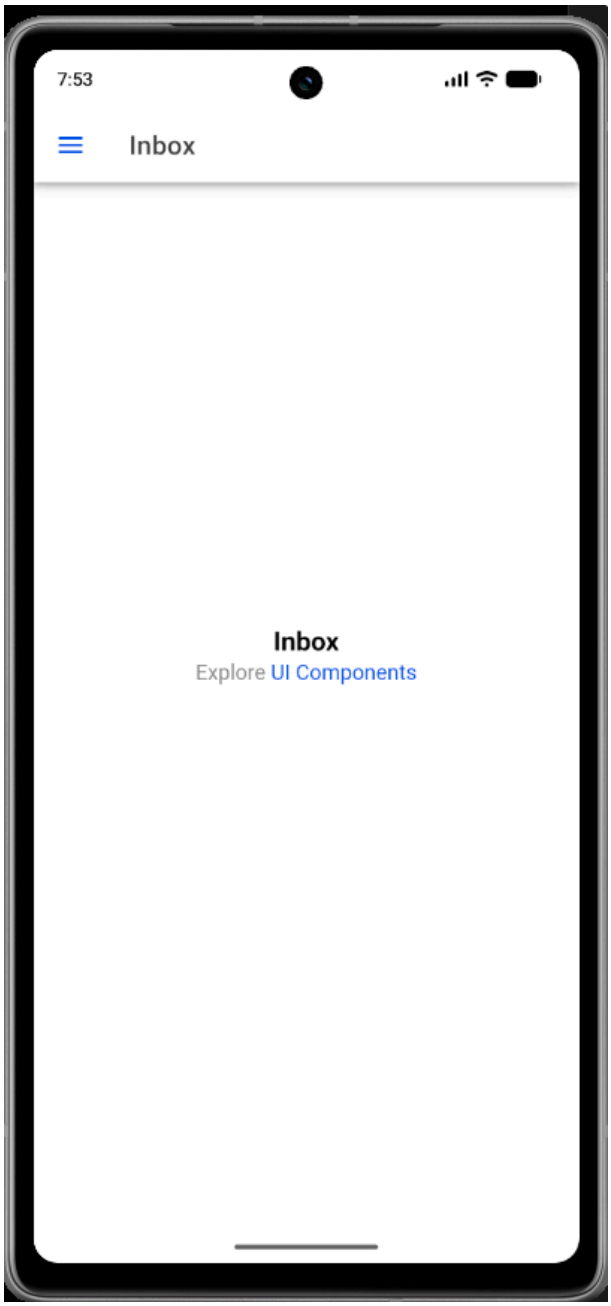
        Re-running this command with the --verbose flag may provide more information.
```

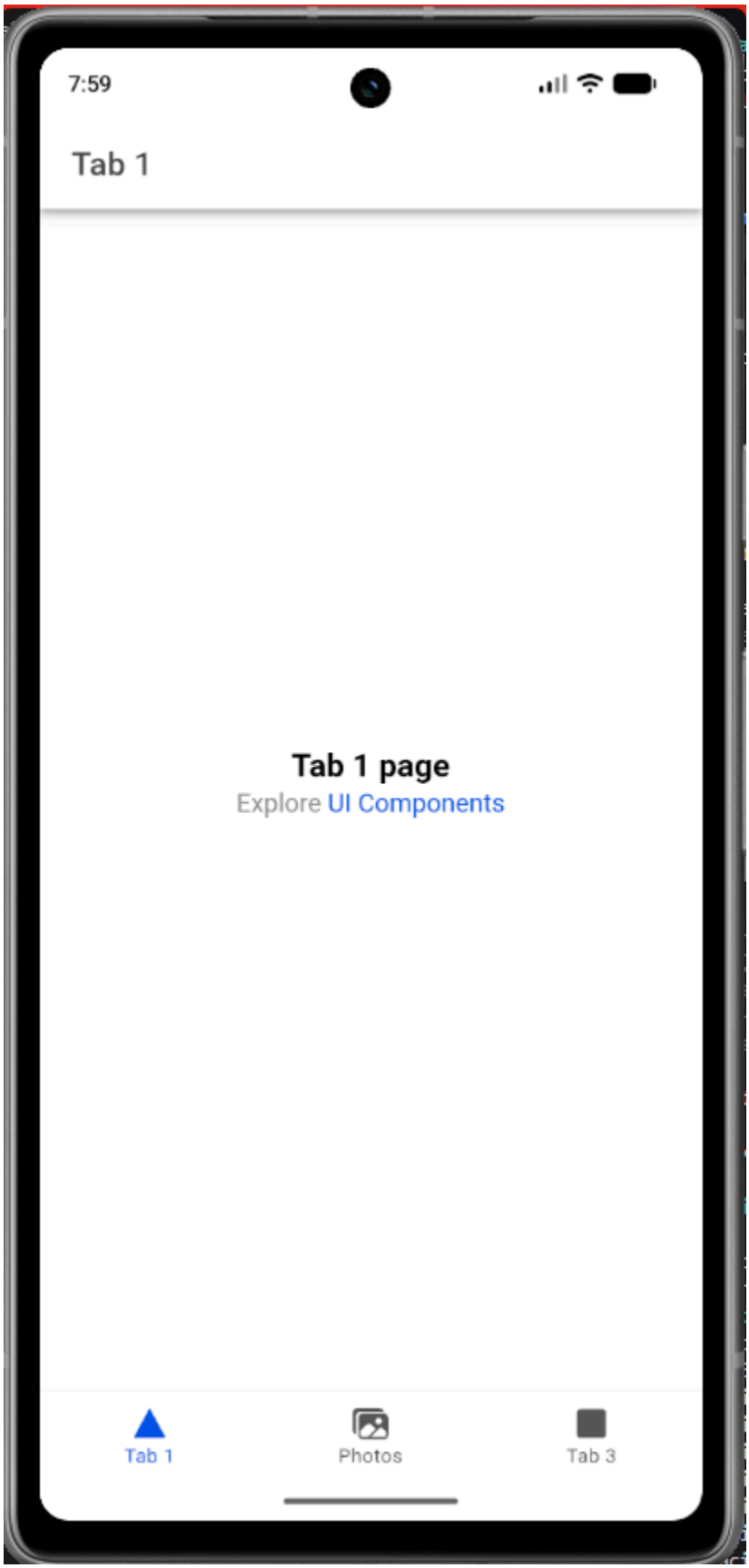
```
PS C:\Users\11_2DAW\Desktop\dah_2627\app-plantilla-list\app-plantilla-list> ionic capacitor run android -l --external
? Which device would you like to target? (Use arrow keys)
> Google sdk_gphone16k_x86_64 (emulator-5554)
  Pixel 7a (emulator) (Pixel_7a)
```

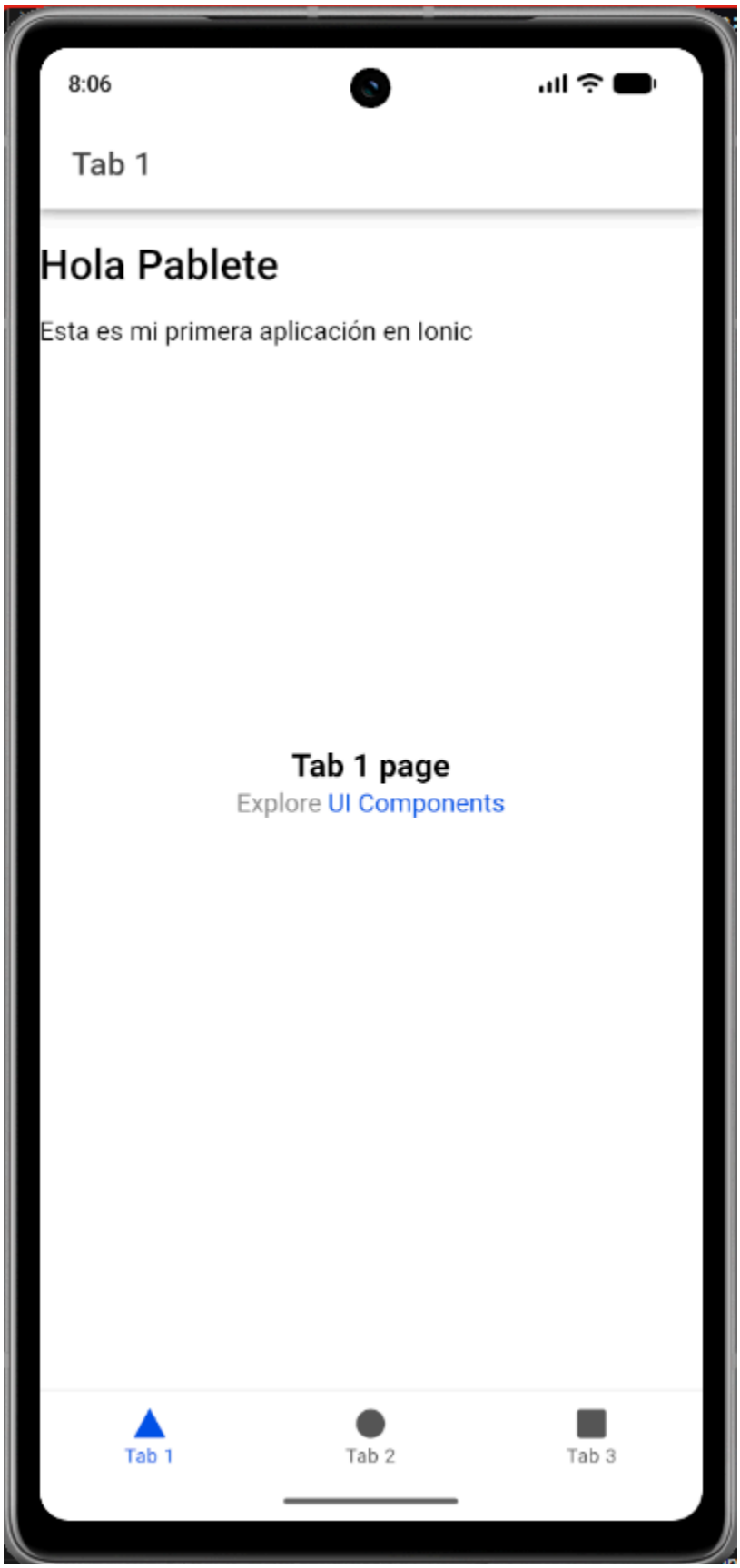
El problema estaba en que la ruta de ANDROID_HOME estaba definida pero no podia leerlo el comando de ionic, asique la he borrado, rehecho y reiniciado el PC, al ejecutar de nuevo he conseguido desplegar la app



Iremos probando a desplegar el resto de aplicaciones


7:53
Inbox
Inbox
Explore UI Components


7:59
Tab 1
Tab 1 page
Explore UI Components
Tab 1
Photos
Tab 3


8:06
Tab 1
Hola Pablete
Esta es mi primera aplicación en Ionic
Tab 1 page
Explore UI Components
Tab 1
Tab 2
Tab 3